Cullen Merritt

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Education

Iowa State University, Ames, IA **3.9/4.0 GPA Cumulative**
B.S. Mechanical Engineering, minor in Cyber Physical Systems Expected Graduation May 2028

Experience

NASA – mTRAX Planetary Exploration Intern | Glenn Research Center, Cleveland OH August 2026 – December 2026

- Own an **independent rigid-wheel test campaign** on the TREC single-wheel dynamometer, measuring net traction and sinkage across forced slip under set vertical loads in two lunar simulants and correlating with contact stress models
- Run tractive performance and durability tests on Lunar Outpost and Astrolab tires for **NASA's Lunar Terrain Vehicle (LTV)**, the crewed rover for **Artemis surface operations**, on TREC and the **Simulated Lunar Operations (SLOPE)** full-vehicle rig, covering instrumentation, test execution, data acquisition, and post-processing

Collins Aerospace – Avionics Manufacturing Intern | Burnsville MN May 2026 – August 2026

- Redesigned a pneumatic die press for platinum tube forming in Creo, adding a self-aligning feature and replacing a moving internal tube support with a stationary one **avoiding $30,000 in vendor fixture procurement**
- Set the CAD origin at the forming contour's datum so the model, drawings, and CMM inspection shared one reference frame, and resolved a tolerance stack-up gap after first parts missed tolerance
- Improved a wind tunnel fixture design, **reducing operator touch time by 23% (6 min per cycle)**

BAE Systems Inc – Technical Operations Manufacturing Co-op | Cedar Rapids IA May 2025 – December 2025

- Designed a burn-in test fixture with a quarter turn spring clamp that sets heat sink contact force by geometry instead of operator torque without obstructing convective airflow, sizing the spring with beam theory and FEA
- After an early prototype failed burn-in from low contact pressure, instrumented heat sink fins with **16 thermocouples** to baseline the qualified thermal profile and iterated the design against it
- **Cut fixture changeout 70% (45 min to 13 min per rack)** on the throughput limiting step, recovering idle time on burn-in chambers running continuously and **saving ~$75,000/year in labor (time study based)**
- Implemented 5S tote management across 10 product configurations, **reducing excess totes 45%** (**Pioneer Award**)

Projects

Tactus – Wearable Macro-pad for Note Taking January 2026 – Present

- Designed a hands-free, wearable shortcut remote that sends keyboard shortcuts to iPad apps over Bluetooth, with enclosures manufactured in-house via 3D printing (TPU band, PETG-CF housing, PLA case)
- **Designed a custom 20 x 70 mm, 2-layer ESP32-C6 PCB** with LiPo charging and a load sharing power path, validating 5 assembled boards at **1.1 mV FET drop and 4.19 V charge regulation against a 4.20 V spec**
- Built a custom BLE HID firmware (deep sleep with wake on any button, on-device Wi-Fi captive portal remapping without reflash) using AI coding agents; debugged and validated it on hardware

Iowa State Undergraduate Research – Development of Robotic Hand August 2024 – May 2025

- Built the open-source Robot Nano Hand, a 3D-printed, tendon-driven hand with 11 servos and a Jetson Nano
- Implemented Python-based motion control and actuator coordination for multi-DOF articulation

Leadership & Involvement

Iowa State University – Mechanics of Materials (EM 3240) Teaching Assistant | Ames IA January 2026 – May 2026

- Taught statically indeterminate systems, beam deflection, combined loading, Mohr's circle, failure theories, and buckling

Solidworks Club January 2026 – May 2026

Phi Kappa Psi – Scholarship Chair January 2025 – May 2025

- Tracked chapter grades, helped struggling members, ran study sessions

Boxing Club August 2023 – May 2025

Phi Kappa Psi – Chair of Professional Development January 2024 – May 2024

- Ran resume and networking workshops for chapter

The Engineering Ambassador and Mentorship Club January 2024 – December 2025

Technical Skills & Interests

- Design & CAD: SolidWorks, PTC Creo, Siemens NX, ANSYS, GD&T per ASME Y14.5, KiCad
- Test & Measurement: thermocouple instrumentation, data acquisition, CMM inspection, dynamometer testing
- Manufacturing & Fabrication: FDM and resin 3D printing, CNC, fixture design, 5S/Lean
- Programming: Python, MATLAB, VBA, C++
- Cyber Physical Systems: sensors/actuation, controls & embedded integration
- Interests: Grandma's and Twin Cities Marathons (2025), Minnesota State Hockey Tournament (2023)